

1. Traceroute uses three test cases because it is possible that a packet has some kind of a delay at a hop preventing it from reaching its destination quickly or at all. By doing it three times, we are able to see a more accurate time delay at the hop.

2.

```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvements! http

PS C:\Users\erinm> tracert cs.merrimack.edu

Tracing route to cs.merrimack.edu [50.220.237.45]
over a maximum of 30 hops:

  1    9 ms    5 ms    5 ms  10.2.144.1
  2   19 ms   28 ms   30 ms  pix.merrimack.edu [10.0.0.50]
  3    7 ms    6 ms    5 ms  50.220.237.45

Trace complete.
PS C:\Users\erinm> tracert merrimack.edu

Tracing route to merrimack.edu [10.0.0.4]
over a maximum of 30 hops:

  1    *      *      *      Request timed out.
  2    *      *      *      Request timed out.
  3    *      *      *      Request timed out.
  4    *      *      *      Request timed out.
  5    *      *      *      Request timed out.
  6    *      *      *      Request timed out.
  7    *      *      *      Request timed out.
  8    *      *      *      Request timed out.
  9    *      *      *      Request timed out.
 10   *      *      *      Request timed out.
 11   *      *      *      Request timed out.
 12   *      *      *      Request timed out.
 13   *      *      *      Request timed out.
 14   *      *      *      Request timed out.
 15   *      *      *      Request timed out.
 16   *      *      *      Request timed out.
 17   *      *      *      Request timed out.
 18   *      *      *      Request timed out.
 19   *      *      *      Request timed out.
 20   *      *      *      Request timed out.
 21   *      *      *      Request timed out.
 22   *      *      *      Request timed out.
 23   *      *      *      Request timed out.
 24   *      *      *      Request timed out.
```

C. The machine hosting merrimack.edu resides off campus because after the packet reaches the gateway it leaves and never comes back unlike cs.merrimack.edu which leaves through the gateway and then immediately enters the Merrimack network again.

D. The route is through the edge network because when it leaves through the gateway, it returns through the Merrimack network shown in the private IP address for pix.merrimack.edu.

3.

```

Trace complete.
PS C:\Users\erinm> ping /n 10 cs.merrimack.edu

Pinging cs.merrimack.edu [50.220.237.45] with 32 bytes of data:
Reply from 50.220.237.45: bytes=32 time=9ms TTL=62
Reply from 50.220.237.45: bytes=32 time=9ms TTL=62
Reply from 50.220.237.45: bytes=32 time=7ms TTL=62
Reply from 50.220.237.45: bytes=32 time=7ms TTL=62
Reply from 50.220.237.45: bytes=32 time=8ms TTL=62
Reply from 50.220.237.45: bytes=32 time=8ms TTL=62
Reply from 50.220.237.45: bytes=32 time=7ms TTL=62
Reply from 50.220.237.45: bytes=32 time=15ms TTL=62
Reply from 50.220.237.45: bytes=32 time=8ms TTL=62
Reply from 50.220.237.45: bytes=32 time=9ms TTL=62

Ping statistics for 50.220.237.45:
    Packets: Sent = 10, Received = 10, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 7ms, Maximum = 15ms, Average = 8ms

```

```
PING cs.merrimack.edu (50.220.237.45) 56(84) bytes of data.
```

```
--- cs.merrimack.edu ping statistics ---
```

```
3 packets transmitted, 0 received, 100% packet loss, time 2056ms
```

C. When using the terminal to run the ping command all 10 packets that were sent were received while using an online ping, 3 packers were sent and all 3 of them were lost. This means that the server must be a private server that can only be accessed by other devices on the same network.

4. $30\text{m}/3 \times 10^8\text{m/s}$ (not complete here)
- 5.